LANGMUIR

pubs.acs.org/Langmuir Article

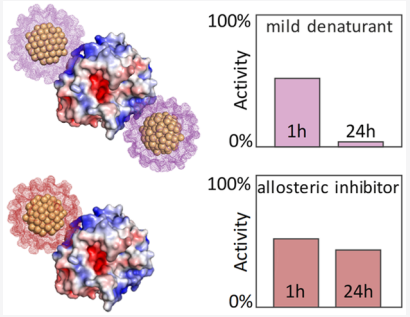
Regulation of Thrombin Activity with Ultrasmall Nanoparticles: Effects of Surface Chemistry

André L. Lira, Rodrigo S. Ferreira, Maria Luiza V. Oliva, and Alioscka A. Sousa*

Cite This: *Langmuir* 2020, 36, 7991−8001 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information

ABSTRACT: Nanomaterials displaying well-tailored sizes and surface chemistries can provide novel ways with which to modulate the structure and function of enzymes. Recently, we showed that gold nanoparticles (AuNPs) in the ultrasmall size regime could perform as allosteric effectors inducing partial inhibition of thrombin activity. We now find that the nature of the AuNP surface chemistry controls the interactions to the anion-binding exosites 1 and 2 on the surface of thrombin, the allosterically induced changes to the active-site conformation, and, by extension, the enzymatic activity. Ultrasmall AuNPs passivated with *p*-mercaptobenzoic acid ligands (AuMBA) and a peptide-based (Ac-ECYN) biomimetic coat (AuECYN) were utilized in our investigations. Remarkably, we found that while AuMBA binds to exosites 1 and 2, AuECYN interacts primarily with exosite 2. It was further established that AuMBA behaves as a "mild denaturant" of thrombin leading to catalytic dysfunction over time. Conversely, AuECYN resembles a proper allosteric effector leading to partial and reversible inhibition of the activity. Collectively, our findings reveal how the distinct binding modes of different AuNP types may uniquely influence thrombin structure and catalysis. The present study further contributes to our understanding of how synthetic nanomaterials could be exploited in the allosteric regulation of enzymes.



INTRODUCTION

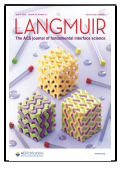
Synthetic nanoparticles (NPs) provide interesting new possibilities toward the regulation of protein structure and function.[1−4] In particular, the use of NPs to regulate the catalytic properties of enzymes is an important area of research.[5,6] Enzymes are integral players within the complex network of biological interactions; hence, control over enzyme activity with exogenous effectors offers a powerful strategy to understand molecular pathways and treat diseases.[7]

NPs can modulate enzyme catalysis in several ways.[8−26] For example, NPs that bind around the active site can influence the activity by regulating substrate capture and product release.[16] NPs can also serve as platforms for the covalent immobilization of enzymes, leading to NP−enzyme hybrids with enhanced stability and activity compared to free enzymes in solution.[10,27−29] As more recently demonstrated, NPs are also able to affect enzyme function by allosteric means.[17,26] By definition, this involves changes in substrate binding affinity and/or catalysis at the orthosteric site in response to NP binding to allosteric (distal) sites. The regulation of enzyme activity via allosteric mechanisms is unique in that it allows a whole range of responses to be obtained, from partial to full inhibition or activation.[30,31] However, to perform as suitable allosteric effectors, NPs must exert the proper kind of conformational and/or dynamic changes to the active site, while substantial perturbation to the overall folded state of the protein must be avoided.[32]

Ultrasmall gold NPs (AuNPs) and nanoclusters represent an attractive class of nanomaterials for use as allosteric modulators of enzyme activity.[17,26,32] Being smaller than most proteins, ultrasmall AuNPs can target specific regions on a protein surface through a limited binding interface.[1,17,33] Detailed kinetics characterization has further revealed that the interactions of ultrasmall AuNPs with proteins are short-lived, displaying residence times for complex dissociation in the range of seconds.[34−36] Drawing from these favorable properties, we recently showed that *p*-mercaptobenzoic acid (*p*MBA)-coated AuNPs of 2 nm in core diameter (AuMBA) could perform as synthetic receptors to allosterically regulate thrombin activity. Specifically, we found that the negatively charged AuMBA particles interacted selectively with thrombin's positively charged exosites via electrostatic forces, producing long-range conformational changes to the active site and partial inhibition of the enzymatic activity.[17]

In the present work, we focus on the long-term impact of ultrasmall AuNP interactions on the structure and function of α-thrombin, used here as a model protein. Thrombin is a highly plastic enzyme displaying allosteric linkage between its catalytic site and exosites;[30,37,38] moreover, the active site can be conveniently derivatized with fluorescent probes enabling

Received: May 7, 2020
Revised: June 16, 2020
Published: June 26, 2020



Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

ACS Publications © 2020 American Chemical Society

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001

the characterization of biomolecular interactions using fluorescence methods.[30,39] We find that the nature of the AuNP surface chemistry controls the interactions with the exosites and, by extension, the enzymatic activity. This study furthers our understanding of how nanomaterials (ultrasmall AuNPs in particular) can be exploited in the allosteric control of enzymes.

MATERIALS AND METHODS

Materials and Reagents. Human α-thrombin, fluorescein (FITC) conjugated to the active-site inhibitor Phe-Pro-Arg-Chloromethylketone (PPACK), and the chromogenic substrate S-2238 were obtained from Molecular Innovations (Novi, MI). Biotinylated thrombin, the fluorogenic substrate SN-59, and dansylarginine *N*-(3-ethyl-1,5-pentanediyl)amide (DAPA) were obtained from Haematologic Technologies (Essex Junction, VT). Human fibrinogen and antithrombin III (AT) were obtained from Millipore Sigma (Burlington, MA). The aptamers HD1 and HD22 were synthesized by Exxtend (Campinas, SP). The peptide Ac-ECYN was synthesized by Proteimax (São Paulo, SP). Sensor chip SA with preimmobilized streptavidin was obtained from GE Healthcare Life Sciences (Piscataway, NJ). The remaining reagents were obtained from Sigma-Aldrich (Milwaukee, WI). Thrombin was labeled stoichiometrically with FITC-PPACK as reported previously.[17] The AuMBA nanoparticles were synthesized as described in detail in several previous reports (e.g., see refs 40 and 41). AuECYN was prepared by ligand exchange of AuMBA with Ac-ECYN in 20 mM phosphate buffer using an Ac-ECYN:AuMBA feed ratio of 1000:1. The reaction was performed at room temperature for 2 h under constant agitation. Excess reagent was removed by precipitation in the presence of methanol and ammonium acetate as described before.[40] The AuNPs' size and uniformity were assessed by analytical ultracentrifugation as previously reported.[40] Experiments were performed in either 20 mM phosphate buffer (PB) solution (pH 7.4) or 20 mM Tris-HCl buffer solution (pH 7.4) supplemented with NaCl (typically 0.15 or 0.7 M) and polyethylene glycol 8000 (0.1%), as indicated in the text.

Fluorescence-Quenching Titration. Fluorescence measurements were performed on a Shimadzu spectrofluorimeter model RF-6000. Fluorescence-quenching titration experiments were carried out with an excitation wavelength of 495 and 5 nm bandpass. FITC-labeled thrombin (0.1 μM) was titrated with the AuNPs in PB containing 0.15 M NaCl, and the recorded emission intensities were plotted as a function of the nanoparticle concentration. The inner-filter effect was corrected for by titrating free FITC with identical concentrations of AuNPs and then dividing the uncorrected intensities by the FITC reference.[42,43] Fluorescence-quenching titration in the presence of the exosite-directed aptamers was performed by first equilibrating FITC−thrombin with HD1 (15 μM) and/or HD22 (5 μM) and then proceeding as described above. Fluorescence recovery experiments were performed by pre-equilibrating FITC−thrombin (0.1 μM) with the AuNPs (0.2 μM) for either 0.5 or 24 h, followed by titration with the aptamers. The titrations were performed with one aptamer type first (HD1 or HD22) up to a maximum concentration of 30 μM, followed by titration with the second aptamer (HD22 or HD1) totaling 60 μM of added aptamers. Emission intensities at 520 nm were recorded ~1 min after each titration point and plotted normalized from 0 to 1 for ease of comparison.

Conformational Changes in Thrombin. For the assessment of global structural changes in thrombin, the protein (0.1 μM) was preincubated with the AuNPs (1 μM) for either 0.5 or 24 h in PB containing 0.15 M NaCl. The corresponding emission spectra of tryptophan were recorded using an excitation wavelength of 280 nm and compared to the spectra of free thrombin obtained under identical conditions. CD spectroscopy characterization of the thrombin secondary structure was performed on a Chirascan Plus instrument from Applied Photophysics (Leatherhead, U.K.). Additional details about the CD measurements are presented in the Supporting Information Figure S5 caption. The occurrence of conformational changes limited to the active-site cleft was investigated by incubating FITC−thrombin (0.1 μM) with the AuNPs (0.2 μM) for either 0.5 or 24 h in PB containing 0.15 M NaCl, followed by an increase in the salt concentration of the solution to 0.7 M NaCl. Emission spectra of FITC−thrombin were recorded immediately before and after the increase in salt concentration. An identical procedure was conducted with FITC−thrombin in the absence of AuNPs as the control. In another set of experiments, FITC−thrombin (0.1 μM) was titrated with the AuNPs in the absence or presence of HD1 (15 μM), HD22 (5 μM), and HD1 + HD22 in PB solution containing 0.15 M NaCl. The recorded emission spectra and wavelengths of maximum emission were plotted as a function of the nanoparticle concentration; a skewed Gaussian function was used in the determination of the wavelengths of maximum emission.[44] Fluorescence anisotropy measurements were carried out in a Chirascan Plus instrument. FITC−thrombin (0.1 μM) was preincubated with the AuNPs (0.2 μM) for either 0.5 or 24 h in PB containing 0.15 M NaCl, followed by an increase in the salt concentration to 0.7 M NaCl. Anisotropy values were recorded immediately before and after the increase in salt concentration. The excitation wavelength was set to 495 nm, and the emission signal was recorded with a 515 nm long-pass filter. An identical procedure was conducted with FITC−thrombin in the absence of AuNPs as the control.

Surface Plasmon Resonance Spectroscopy (SPR). SPR measurements were performed in a Biacore T-200 instrument from GE Healthcare Life Sciences (Pittsburgh, PA). Biotinylated thrombin (150 response units, RU) was immobilized onto a commercial streptavidin-coated sensor surface. PB solution containing 0.15 M NaCl was used as a running buffer. The AuNP solutions were injected in the flow at a rate of 90 μL/min in the concentration range from 1 nM to 20 μM. The surface was regenerated between injections with 0.005% sodium dodecyl sulfate in water followed by 2 M NaCl. A reference, unmodified surface was used for the background subtraction and correction of refractive index changes. The SPR data were analyzed with the software EVILFIT assuming that the surface sites represent a continuous distribution of equilibrium and kinetic rate constants.[45,46]

Stopped-Flow Spectroscopy. Stopped-flow measurements were performed in an SX20 instrument from Applied Photophysics equipped with a 490 nm LED light source and a 515 nm long-pass filter at the excitation path. Samples were prepared in PB solution containing 0.15 NaCl and supplemented with 20 μM HD22. The association reactions were monitored at 6 °C using FITC−thrombin and AuNP concentrations of 0.1 and 1 μM, respectively. The kinetics of dissociation were followed by competitive displacement of preformed complexes of FITC−thrombin/AuNPs (0.1 μM/1 μM) with excess unlabeled thrombin (20 μM). All given values of concentration are after mixing.

Thrombin Recognition of Physiological Substrates. Thrombin (0.5 nM) was incubated with the AuNPs (0.5 μM) for either 0.5 or 24 h in PB containing 0.15 M NaCl, followed by the addition of fibrinogen (2 μM). In another set of experiments, AT/heparin (2 μM/100 μM) was added to the mix immediately before the addition of fibrinogen. The kinetics of fibril formation upon the thrombin-dependent activation of fibrinogen were monitored by recording absorbance measurements at 350 nm as a function of time.

Enzyme Activity Assays. The experiments were performed in a Spectramax instrument from Molecular Devices (Sunnyvale, CA) at 25 °C. Tris-HCl (20 mM, pH 7.4) containing 0.15 M NaCl was used as a buffer solution. The concentration-dependent inhibition of thrombin by the AuNPs was investigated by titrating the particles in a 96-well plate containing a fixed amount of thrombin, followed by the addition of S-2238 to each well simultaneously. The final concentration of reagents in the wells was 2 nM for thrombin, 150 μM for S-2238, and 0.004, 0.008, 0.016, 0.03, 0.063, 0.13, 0.25, 0.5, 1, and 2 μM for the AuNPs. Control samples for background subtraction were prepared in a similar way but in the absence of thrombin. Initial velocities (v_0) were measured from the slope of the initial (linear) part

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001

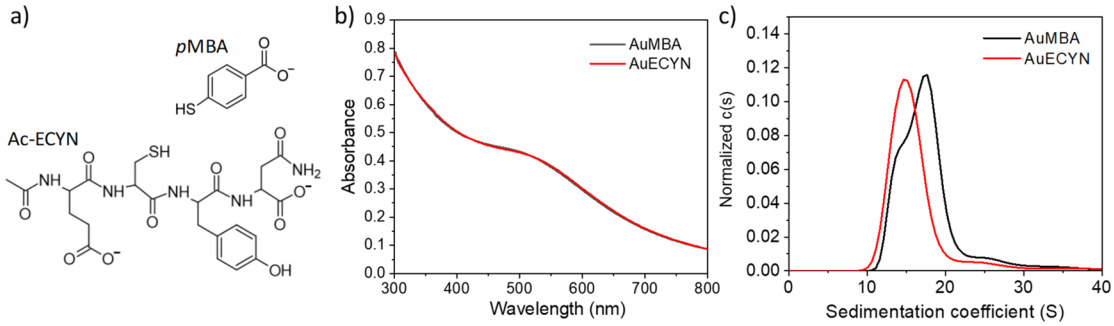


Figure 1. Characterization of AuMBA and AuECYN. (a) Structures of the *p*-mercaptobenzoic acid (*p*MBA) and Ac-ECYN peptide ligands. The carboxylic acid groups are deprotonated at pH 7.4, hence imparting both ligands with a net negative charge. (b) UV−visible spectra of the AuNPs. (c) Analytical ultracentrifugation analysis of the AuNPs.

of the progress curves (absorbance readings over time). Percent activities were computed relative to the initial velocities obtained from control samples lacking AuNPs. The influence of the incubation time with AuNPs on thrombin activity was investigated by first mixing together thrombin (2 nM) and nanoparticles (0.5 μM) for either 0.5 or 24 h, followed by the addition of substrate. A separate set of samples was prepared in a similar way, except that the NaCl concentration was increased to 0.7 M immediately before the addition of substrate. Corresponding vales of v_0 were recorded at 0.15 and 0.7 M NaCl and normalized to the v_0 obtained in the absence of particles.

RESULTS AND DISCUSSION

The data presented in Supporting Information Figure S1 summarize the influence of AuMBA interactions on the structure and function of thrombin.[17] Briefly, AuMBA binds selectively to exosites 1 and 2 with an apparent affinity constant (K_D) of ∼40 nM. AuMBA complexation with thrombin leads to allosteric reconfiguration of the active site and allosteric modulation of the substrate binding affinity and catalysis. The enzymatic activity is found to be partly reduced—but not completely inhibited—by saturating concentrations of AuMBA.

However, later experiments showed that extending the incubation time with AuMBA for 24 h produced increased conformational changes in the active site and a complete and irreversible inhibition of the enzyme activity (vide infra). We hypothesized that the time-dependent inactivation of thrombin by ultrasmall AuMBA could be averted by redesigning the nanoparticle surface chemistry. We therefore prepared and tested a new ultrasmall AuNP incorporating a peptide-based biomimetic coat formed by Ac-ECYN. In the following sections, we describe the preparation of Ac-ECYN-coated ultrasmall AuNPs (AuECYN) and present a systematic side-by-side comparison of the interactions of AuMBA and AuECYN with thrombin. All reported data were recorded under identical experimental conditions to enable an accurate comparison between the two nanoparticle types.

Synthesis and Characterization of Ultrasmall AuNPs Passivated with Ac-ECYN. AuECYN was prepared by ligand exchange of AuMBA with Ac-ECYN. We drew inspiration from our previous studies with glutathione-coated ultrasmall AuNPs to design the tetrapeptide Ac-ECYN employed in this work.[34,35,40,47,48] Ac-ECYN contains a cysteine residue for binding to Au, a glutamate residue to increase the ligand cone angle and prevent nanoparticle etching during ligand exchange,[47] a tyrosine residue capable of establishing both polar and nonpolar interactions across the binding interface, and an asparagine residue to impart a highly polar environment at the outermost portion of the nanoparticle coating. Moreover, the glutamate residue is acetylated in order to neutralize the positive charge at the N-terminal amino group since the presence of this charge would decrease the binding affinity toward thrombin. Of note, the carboxylic acid groups in both *p*MBA and Ac-ECYN are deprotonated at pH 7.4, hence imparting both ligands with a net negative charge (Figure 1a). Characterization data for AuMBA and AuECYN are shown in Figures 1b and 1c. The UV−visible spectra of the AuNPs show the presence of a shoulder but lack of a distinct surface plasmon resonance peak near 500 nm, consistent with particles of ∼2 nm in core diameter (Figure 1b). In addition, the similarity between the spectral profiles of AuMBA and AuECYN imply that the particles have virtually identical core sizes. Analytical ultracentrifugation analysis of AuMBA revealed a distribution of sedimentation coefficients(s) around ∼18 S, again consistent with particles of ∼2 nm in average core diameter (Figure 1c).[40] For AuECYN, the slight shift of the sedimentation rate to lower s-values can be explained by increased hydrodynamic friction of the particles due to the Ac-ECYN coat. For both AuNPs, the spread in s-values from ∼10 to 30 S is consistent with a distribution of nanoparticle sizes from ∼1.5 to 2.5 nm.[40] In conclusion, it can be seen that the AuNPs display similar average sizes and size distributions, so that they differ almost exclusively on the basis of the chemical identity of their passivating layers.

AuMBA and AuECYN Interactions with Thrombin. Similarly as described in our previous work, the surface plasmon resonance (SPR) spectroscopy technique was utilized to characterize the AuNP−thrombin interactions quantitatively.[35] SPR is prone to artifacts, and the results must be interpreted with caution. For example, we have previously observed that thrombin must be immobilized at low enough densities to minimize the occurrence of mass transport limitation and avidity effects.[35] Here, 150 response units (RU) of thrombin were immobilized to a precoated streptavidin sensor surface via a biotin label attached to the active site. This immobilization strategy keeps both exosites exposed to allow interactions with the AuNPs. Quantification of the binding parameters (binding affinity and kinetic rate constants) was accomplished by analysis of the SPR data with a continuous surface-site distribution model.[45,46] For AuMBA, the calculations yielded a single apparent binding affinity (K_D) of ∼40 nM, whereas for AuECYN the apparent K_D was ∼130 nM (Figure S2). The similarity between the apparent affinity constants and their magnitude in the low nM range are both in qualitative agreement with the results from fluorescence-

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001

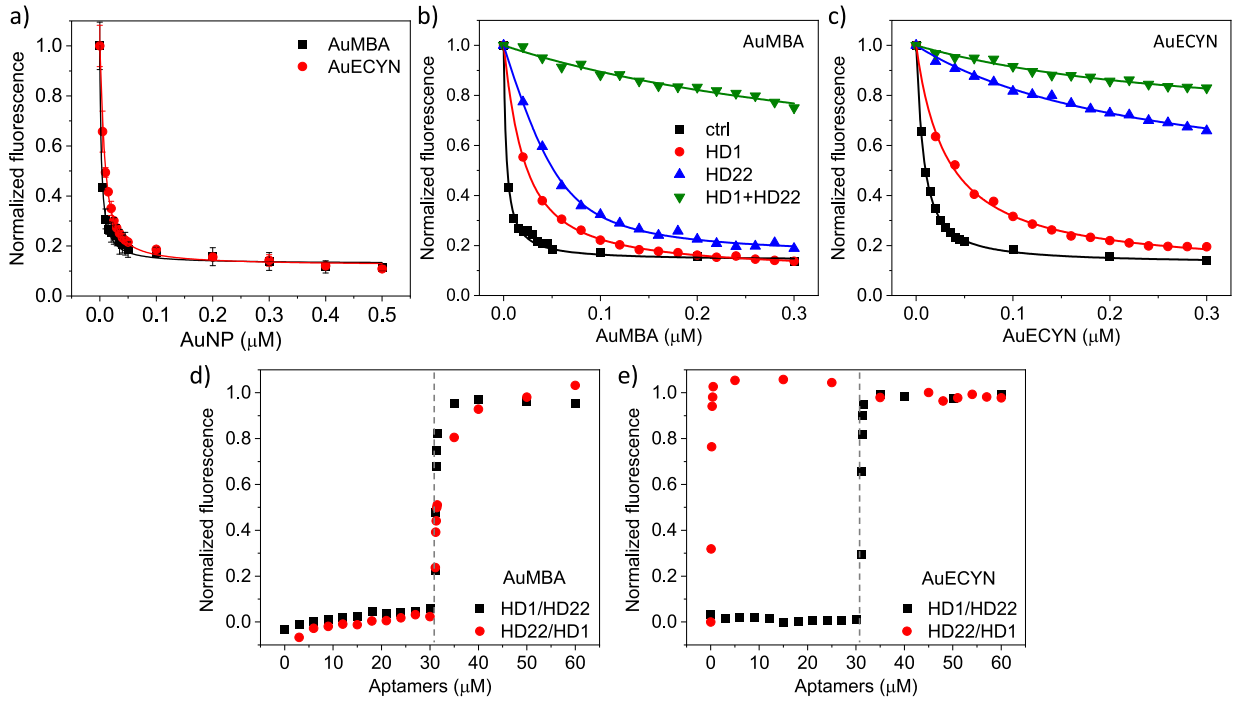


Figure 2. Characterization of AuNP−thrombin interactions by fluorescence spectroscopy. (a) Fluorescence-quenching titration of FITC-labeled thrombin (0.1 μM) with the AuNPs. (b, c) Fluorescence quenching titration of FITC−thrombin (0.1 μM) with (b) AuMBA and (c) AuECYN in the absence (ctrl) or presence of the aptamers HD1 (15 μM), HD22 (5 μM), and HD1 + HD22. Lines are a guide to the eye. Line colors in (c) are the same as in (b). (d, e) Competitive displacement of (d) AuMBA and (e) AuECYN from preformed complexes with FITC−thrombin by HD1 and HD22. FITC−thrombin (0.1 μM) was preincubated with the AuNPs (0.2 μM) for 0.5 h followed by titration with HD1/HD22 or HD22/HD1. The dashed line marks the point at which titration of the first aptamer ends and that of the second aptamer begins. Experiments were performed in phosphate buffer solution supplemented with 0.15 M NaCl.

quenching titration (Figure 2a; vide infra). The corresponding apparent dissociation rate constants (k_{off}) were also very similar between AuMBA and AuECYN: ~0.13 and 0.21 s^{-1}, respectively. These relatively large values of k_{off} indicate that the AuNPs form rather short-lived complexes with thrombin, with apparent residence times of ~8 and 5 s for AuMBA and AuECYN, respectively.

The susceptibility of the interactions to ionic strength was assessed by fluorescence spectroscopy. The results confirmed the electrostatic nature of AuMBA and AuECYN binding to thrombin (Figure S3). Fluorescence-quenching titration of FITC-labeled thrombin with the AuNPs was employed next to characterize the interactions in more detail (Figure 2a). Proper quantification of fluorescence quench data from a single titration curve is not possible due to the multisite nature of complexation.[42,43] Nevertheless, the similarity of the fluorescence decay profiles obtained for AuMBA and AuECYN implies similar apparent binding affinities, in agreement with the SPR results. Titrations were also performed in the presence of a high concentration of the exosite-directed aptamers HD1 and HD22, which bind specifically to exosites 1 and 2, respectively.[49] The results showed that AuECYN complexation with thrombin is impaired in the presence of HD1 + HD22, thereby confirming the exosites as the main binding regions for AuECYN, as found before for AuMBA[17] (Figures 2b and 2c, green traces).

We next carried out an additional set of titration experiments keeping a fixed high concentration of either HD1 or HD22 in order to monitor the interactions of the AuNPs with each exosite alone. We found that AuECYN complexation with exosite 1 (Figure 2c, blue trace) produces a relatively modest drop in fluorescence intensity when compared to the large signal decay obtained from the AuECYN association with exosite 2 (Figure 2c, red trace). This implies that AuECYN binding to exosite 1 is noticeably weaker than binding to exosite 2. In contrast, AuMBA binds with a similarly high affinity toward both exosites (Figure 2b, blue and red traces).[17] Similar conclusions could be derived from a set of complementary experiments whereby preincubated mixtures of thrombin and AuNPs were titrated with HD1 and HD22 (Figures 2d and 2e). For AuMBA−thrombin mixtures, titration with either HD1 or HD22 displaces AuMBA from the corresponding exosite; however, this is not accompanied by an increase in the fluorescence signal because AuMBA remains bound to thrombin via the other exosite. Titration with the second aptamer dislodges AuMBA from the second exosite, and as a result, the fluorescence emission increases sharply (Figure 2d).[17] For AuECYN−thrombin mixtures on the other hand, titration with HD22 maximally increases the fluorescence signal, and further titration of HD1 does not generate additional fluorescence recovery (Figure 2e). This again suggests that AuECYN interactions with exosite 1 are much weaker than with exosite 2.

It is difficult to ascertain the exact molecular mechanism responsible for the different affinities of AuMBA and AuECYN toward exosite 1. In any event, it is interesting to note that AuECYN incorporates a more complex surface architecture that is presumably more solvated. Solvation properties, in turn, are well-known to influence the biomolecular interactions of proteins and NPs alike.[34,50−55] Interactions between hydrophilic and loosely packed protein surfaces, in particular, may be strongly mediated by bridging water molecules forming a

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001

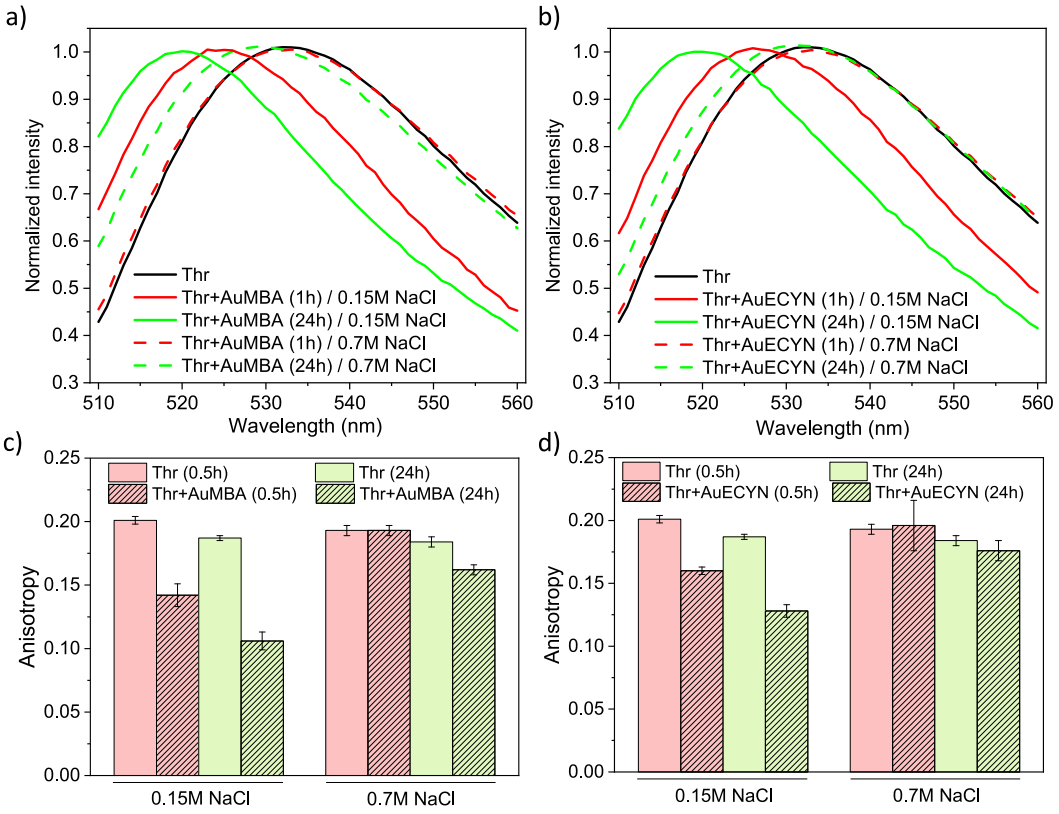


Figure 3. Conformational changes to thrombin's active site. (a) Influence of AuMBA interactions on the fluorescence emission spectra of FITC-labeled thrombin. Thrombin was preincubated with AuMBA for either 0.5 or 24 h in 0.15 M NaCl, followed by an increase in the NaCl concentration to 0.7 M. Fluorescence spectra were recorded immediately before and after the increase in NaCl concentration. (b) Same as in (a), except that AuECYN was used. A permanent blue-shift of the FITC emission signal (measured from the difference between the black and green dashed traces) is apparent after 24 h incubation with AuMBA ($\Delta\lambda \sim 3.1$ nm) and AuECYN ($\Delta\lambda \sim 1.2$ nm). (c) Influence of AuMBA interactions on the fluorescence anisotropy of FITC-labeled thrombin. Thrombin was preincubated with AuMBA for either 0.5 or 24 h in 0.15 M NaCl, followed by an increase in the NaCl concentration to 0.7 M. Anisotropy measurements were recorded immediately before and after the increase in NaCl concentration. (d) Same as in (c), except that AuECYN was used. Error bars are standard deviations from triplicate measurements. FITC–thrombin, 0.1 μM; AuNPs, 0.2 μM.

hydrogen-bond network across the interface.[54,55] A water-rich interfacial gap is also expected in ultrasmall AuNP–protein complexes that are held together primarily by long-range electrostatic forces across a binding interface of low geometric complementarity. We therefore performed stopped-flow kinetic measurements of AuMBA– and AuECYN–thrombin complexation to try and gain further mechanistic insights into the interactions.[35] The experiments were done in the presence of a high concentration of HD22 in order to block exosite 2 and thereby monitor the interactions to exosite 1 alone (Figure S4a). Qualitatively, the results indicated a smaller apparent rate constant of association for AuECYN binding to exosite 1 in relation to AuMBA (Figure S4b). It is reasonable to speculate that this difference might stem from a higher energy penalty for displacing water and ions from the surface of AuECYN, as described before for a similar system.[34] The stopped-flow data also revealed a higher apparent rate constant of dissociation of the AuECYN–exosite 1 complex (Figure S4c). Presumably, this might be the result of somewhat greater levels of hydration at the binding interface between AuECYN and exosite 1, thus leading to weaker electrostatic attraction and lower complex stability. Together, the differences in the association and dissociation kinetics for AuMBA vs AuECYN binding to exosite 1 account for the much higher apparent K_D of the AuECYN–exosite 1 complex.

Impact of AuMBA and AuECYN on the Global Structure of Thrombin. We examined the impact of both short- and long-term interactions (0.5 and 24 h, respectively) of the AuNPs on the global conformation of thrombin. First, CD spectroscopy revealed that AuNP binding to thrombin decreases α-helical and increases β-sheet content (Figure S5). Next, we verified that neither AuNP triggers major changes to the protein tertiary structure, even after a 24 h incubation period. One evidence for this comes from measurements of intrinsic tryptophan fluorescence, which reveal no detectable shifts in the emission spectra of tryptophan in the presence of the AuNPs (Figure S6). Second, the AuNPs do not noticeably affect the capacity of macromolecular ligands to interact with thrombin. In particular, Figure 2d, 2e, and S7 show that HD1 and HD22, used as model ligands, are able to effectively displace the AuNPs from the surface of thrombin by binding to the exosites. Additional data showing thrombin recognition of physiological substrates in the presence of the AuNPs is reported below.

Impact of AuMBA and AuECYN on the Active-Site Conformation of Thrombin. The occurrence of conformational changes limited to the active-site cleft was next investigated using FITC as a reporter of environmental changes at this site (Figure S8).[39,56] As shown previously for AuMBA, AuECYN complexation with thrombin produces an

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991–8001

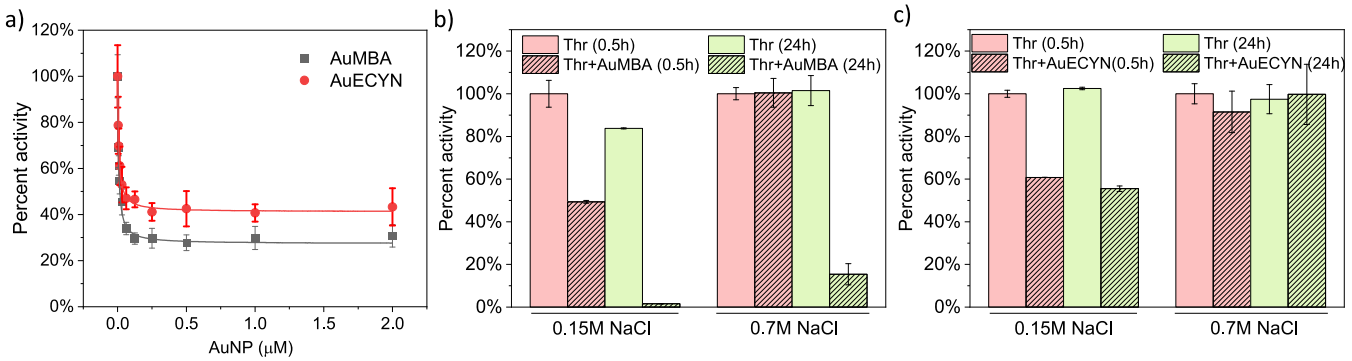


Figure 4. Impact of nanoparticle interactions on thrombin activity. (a) Concentration-dependent inhibition of thrombin (Thr) by AuMBA and AuECYN. Lines are a guide to the eye. (b) Influence of the incubation time with AuMBA on thrombin activity. Thrombin was preincubated with AuMBA for either 0.5 or 24 h in 0.15 M NaCl, followed by the addition of the substrate. A separate set of samples was prepared in a similar way, except that their NaCl concentration was increased to 0.7 M before the addition of substrate. (c) Same as in (b), except that AuECYN was used. Initial velocities were measured from the slope of the initial (linear) part of the progress curves. Percent activities were computed relative to the initial velocities obtained from control samples lacking AuNPs. Error bars are standard deviations from triplicate measurements from three independent experiments. Thrombin, 2 nM; S-2238, 150 μM; AuNPs, 0.5 μM in (b, c). Assays performed in 20 mM Tris-HCl buffer solution at pH 7.4.

instantaneous and substantial blue-shift of the FITC emission spectrum (Figures S8d and S8f, black squares). Interestingly, no spectral shift occurs when AuECYN is titrated in the presence of excess HD22 (Figure S8f, blue triangles), consistent with a lack of major AuECYN binding to exosite 1 as noted above. For AuMBA in comparison, a blue-shift is always observed except when both exosites are blocked simultaneously with HD1 and HD22 (Figure S8c).

Next, the reversibility of the spectral shifts was assessed by incubating thrombin with the AuNPs for 0.5 h in 0.15 M NaCl, followed by an increase in the salt concentration to 0.7 M NaCl. The increase in ionic strength screens the electrostatic interactions and thereby disrupts the AuNP−thrombin complexes (Figure S3). The results showed that the FITC emission spectra return to their original states upon complex dissociation (Figures 3a and 3b, black vs dashed red traces), thus implying that the changes to the active-site conformation are largely reversible. However, experiments performed after a 24 h incubation with the AuNPs revealed an augmented and irreversible blue-shift of the emission spectra (Figures 3a and 3b, black vs dashed green traces). This indicates a reconfiguration of the active-site structure with time by the AuNPs. The amount of irreversible blue-shift after 24 h was measured revealing a slightly larger value for AuMBA relative to AuECYN (~3.1 vs 1.2 nm).

We also performed measurements of the fluorescence anisotropy of FITC-labeled thrombin as another way to monitor the occurrence of conformational changes in the catalytic site. Anisotropy is a sensitive tool to unveil changes in the rotational mobility of a fluorophore resulting from local structural perturbations. Anisotropy is also sensitive to changes in the local dielectric environment of a fluorophore affecting its quantum yield.[57] As shown in Figures 3c and 3d (red bars under 0.15 M NaCl), the fluorescence anisotropy of FITC−thrombin is reduced in the presence of both AuMBA and AuECYN after a 0.5 h incubation, indicating some level of structural perturbation at the active site (in the absence of any perturbation the anisotropy values should have increased as a result of slower tumbling of the larger AuNP−thrombin complex). However, the decrease of the anisotropy is entirely reversed upon dissociation of the complexes with 0.7 M NaCl (Figures 3c and 3d, red bars under 0.7 M NaCl). Additional measurements in 0.15 M NaCl following a 24 h incubation with the AuNPs indicated overall lower anisotropies and thus a more significant perturbation of the active-site structure, especially for AuMBA (Figures 3c and 3d; green bars under 0.15 M NaCl). It can be also noticed that the anisotropy change elicited by AuMBA after 24 h is not entirely reversed after increasing the ionic strength (Figure 3c, green bars under 0.7 M NaCl).

From the above results, it appears that long-term interactions with AuMBA might yield greater and more permanent structural changes to the active-site conformation compared to AuECYN. This view is indeed supported by the enzyme activity assays as described next.

Impact of AuMBA and AuECYN on Thrombin Activity. Here we sought to understand how the nanoparticle-induced perturbation of the active site could influence thrombin activity. First, we recorded the initial rates of hydrolysis (v_0) of the chromogenic substrate S-2238 and plotted the results as a function of the AuNP concentration. The corresponding dose−response profile for AuECYN appears similar to the one for AuMBA as observed previously[17] (Figure 4a). Namely, the residual activity decreases sharply until a nanoparticle concentration of around 0.1 μM (approximately matching the sharp drop in fluorescence intensity as seen in Figure 2a), after which it remains constant at ~40%. The partial, nonzero activity obtained at saturating nanoparticle concentrations is consistent with an allosteric mechanism of inhibition,[58] whereby effector binding at a regulatory (distal) site changes the functional activity at the orthosteric site. That the final residual activities are noticeably different between AuMBA and AuECYN (~30 and 40%, respectively) offers yet another clue that thrombin's active site responds differently to interactions with each nanoparticle type. Interestingly, a similar submaximal inhibitory response has been described recently for small molecule probes and aptamers that bind to thrombin's exosites.[30,59] We also assessed the reversibility of the enzymatic inhibition by screening the interactions with 0.7 M NaCl. The results showed that thrombin activity is fully restored upon complex dissociation (Figures 4b and 4c; compare red bars under 0.15 and 0.7 M NaCl).

Inhibition experiments were next conducted in the presence of excess HD1 or HD22 to achieve selective blocking of either

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001

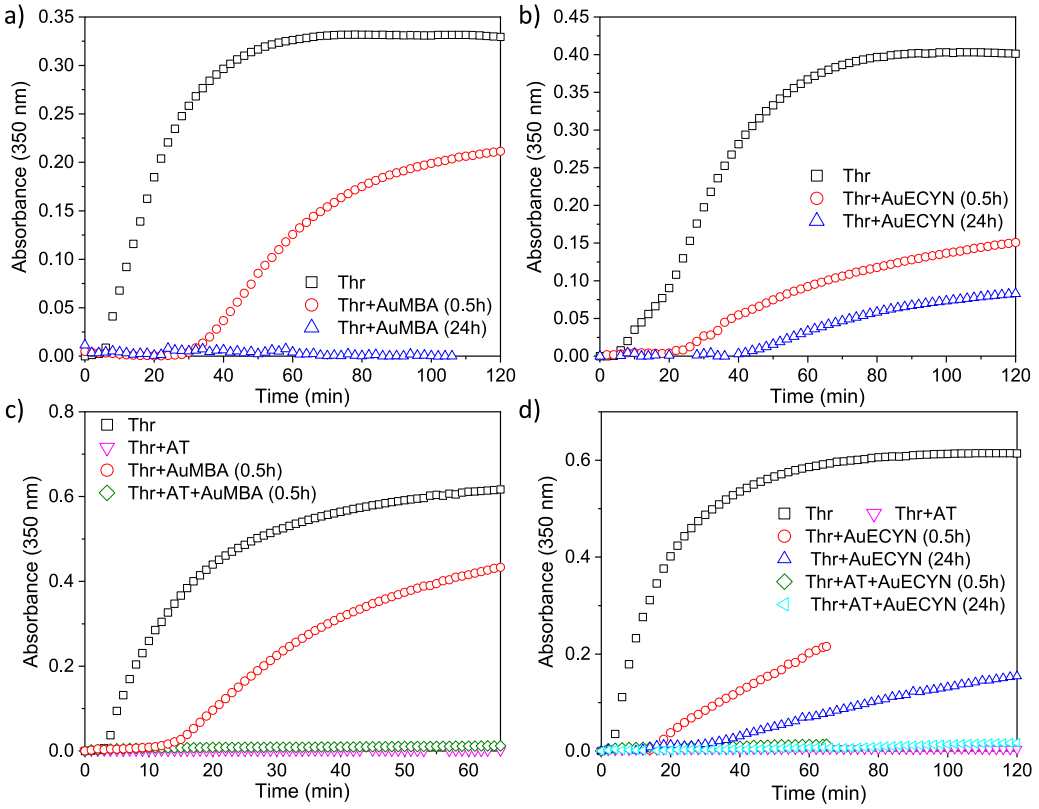


Figure 5. Thrombin recognition of physiological substrates. (a, b) Cleavage of fibrinogen by thrombin in the presence of (a) AuMBA and (b) AuECYN. Thrombin and the AuNPs were preincubated for either 0.5 or 24 h followed by the addition of fibrinogen. Control experiments were performed in the absence of AuNPs. (c, d) Inhibition of the thrombin-dependent cleavage of fibrinogen by antithrombin (AT)/heparin in the presence of (c) AuMBA and (d) AuECYN. Thrombin and the AuNPs were preincubated for either 0.5 or 24 h followed by the sequential addition of AT/heparin and fibrinogen. Control experiments were performed in the absence of AuNPs and AT/heparin as indicated. Experiments were performed in phosphate buffer solution containing 0.15 M NaCl; thrombin, 0.5 nM; AuNPs, 0.5 μM; fibrinogen, 2 μM; AT, 2 μM; heparin, 100 μM.

exosite 1 or 2. The results showed that AuMBA complexation with either exosite alone produces a similar dose−response profile as that which is obtained in the absence of the aptamers (Figure S9a). Incidentally, this confirms that multimeric aggregation cannot be the cause of thrombin inhibition by AuMBA. Rather, inhibition must rely solely on nanoparticle-induced allosteric changes in the active site, since no signs of multimeric aggregation are observed in the ternary system AuMBA/thrombin/aptamer (see ref 17 for details). For AuECYN, we notice reduced levels of thrombin inhibition in the presence of excess HD22 (Figure S9b), in full agreement with the established binding selectivity of AuECYN toward exosite 2. For both AuNP types, their inhibitory effect on thrombin activity was reduced significantly when blocking both exosites simultaneously with HD1 + HD22 (Figure S9).

Additional measurements of enzymatic activity were performed following a 24 h incubation with the AuNPs in 0.15 M NaCl. While the residual activity drops to zero in the presence of AuMBA, it remains approximately constant for AuECYN regardless of the incubation time (Figures 4b and 4c). Similar results are observed when monitoring the hydrolysis of a fluorogenic substrate (SN-59) in a spectrofluorimeter (Figure S10). We also found that the enzymatic inactivation triggered by AuMBA is irreversible; that is, the residual activity remains close to zero upon increasing the solution ionic strength to screen the interactions (Figure 4b, green bars under 0.7 M NaCl). On the other hand, the enzymatic activity is fully recovered with the addition of salt in the case of AuECYN (Figure 4c, green bars under 0.7 M NaCl).

Long-term interactions between thrombin and AuMBA were next investigated in the presence of excess HD1, thereby limiting AuMBA binding to exosite 2 alone (Figure S11). It was found that AuMBA causes reversible inhibition of thrombin activity (but not inactivation), as similarly observed with AuECYN. This indicates that binding of AuMBA to exosites 1 and 2 is required to generate sufficient conformational strain in the active-site region that results in catalytic dysfunction over time. Interestingly, thrombin's active-site cleft remains capable of recognizing small molecular ligands such as the active-site inhibitor DAPA[60] (Figure S12), therefore supporting the view that long-term interactions with AuMBA inactivate the enzyme by disturbing its catalytic—but not ligand-binding—properties.

The irreversible inactivation of thrombin caused by long-term interactions with AuMBA is in line with the structural studies described above, which hinted that the active-site geometry might be more strongly perturbed by AuMBA relative to AuECYN. The disruption of the active-site geometry but lack of major changes to the overall protein conformation is consistent with the prominent conformational flexibility that typically characterizes the active site of enzymes.[61−64] Indeed, many enzymes have been shown to undergo loss of activity under the influence of external factors

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

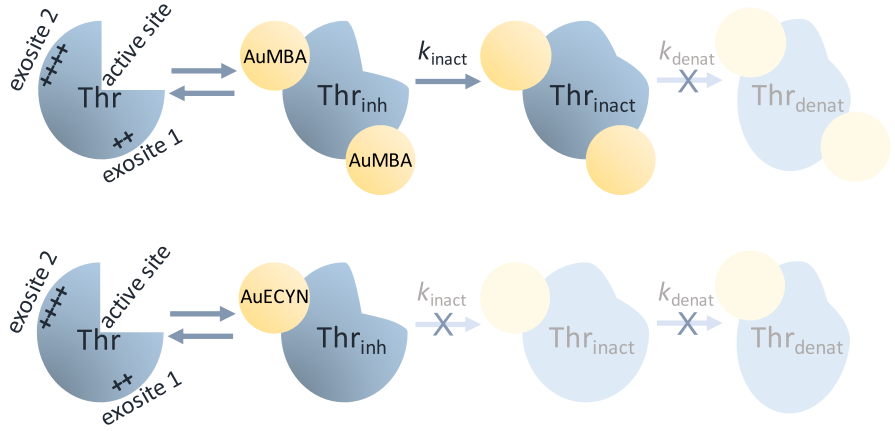


Figure 6. Simplified schematics illustrating the allosteric effects of AuMBA and AuECYN on thrombin structure and catalysis. NP complexation with the exosites drives a partially inhibited conformational state of the enzyme (Thr_{inh}). Upon NP dissociation, the enzyme returns to its native and fully functional form (Thr). Long-term interactions with AuMBA cause the protein to evolve irreversibly to a fully inactivated (but not denatured) state (Thr_{inact}). The same inactivation process leading to Thr_{inact} is not observed with AuECYN (faded Thr_{inact}). Global protein denaturation (Thr_{denat}) is not apparent for either AuMBA or AuECYN under the given experimental conditions (faded Thr_{denat}).

(including denaturants and temperature) before marked changes in overall protein conformation occur.[61,62,64]

In closing this section, we note a previous study reporting thrombin interactions with quantum dots and their effect on enzyme catalysis.[65] Specifically, proteolytic thrombin activity was accelerated when peptide substrates were immobilized onto the surface of quantum dots compared to the substrates in bulk solution. However, the degree of acceleration was strongly dependent on the underlying surface chemistry, with anionic ligand coatings effecting only modest activity enhancement compared to zwitterionic-based coatings. The likely explanation is that thrombin adsorbs more strongly onto the surface of the anionic quantum dots mediated by the positively charged exosites. This previous study offers yet another example of the modulating influence of surface chemistry on thrombin−nanomaterial interactions and, consequently, on thrombin activity.

Impact of AuMBA and AuECYN on the Recognition of Physiological Substrates by Thrombin. We finally examined whether interactions of the AuNPs interfere with the ability of thrombin to form functional complexes with fibrinogen and antithrombin (AT)/heparin. We recall that recognition of fibrinogen by thrombin is mediated by many noncovalent intermolecular forces across an extended binding interface comprising the active site in conjunction with exosite 1.[66] Recognition of AT/heparin similarly entails a complex binding interface with thrombin, including exosite 2, the γ-loop, and the Na^+-binding site.[66] The results described next therefore represent a more stringent test of the impact of NP complexation on thrombin structure and functionality.

The basic experiment consisted of preincubating the AuNPs with thrombin for either 0.5 or 24 h, followed by the addition of fibrinogen. The ability of thrombin to cleave fibrinogen and form a fibrin clot was then monitored by recording absorbance measurements at 350 nm as a function of time. Inhibition experiments involving AT/heparin were performed in a similar way, except that AT/heparin were introduced into the AuNP−thrombin mixtures immediately before the addition of fibrinogen. Representative results for the kinetics of fibril formation are displayed in Figure 5. The kinetic traces are very characteristic and consist of a lag time, sigmodal increase, and plateau.[67] A quantitative analysis of the data shown in Figure 5 is difficult due to the multiple interactions involved, which in turn are dictated by the relative values of binding affinity and concentrations (e.g., fibrinogen and AT will compete with the AuNPs for binding to the exosites). Nevertheless, the results revealed important qualitative differences in behavior between AuMBA and AuECYN as a function of the preincubation time. First, it can be seen that the capacity of thrombin to form functional complexes with fibrinogen and AT/heparin is maintained in the presence of AuMBA provided the incubation time is short (Figures 5a and 5c). On the other hand, thrombin is fully inactivated and thus unable to cleave fibrinogen after 24 h incubation with AuMBA (Figure 5a), this being likely the result of a disturbed active-site conformation as discussed above. In contrast to AuMBA, thrombin remains active toward fibrinogen in the presence of AuECYN regardless of the incubation time (Figure 5b). AuECYN also does not impair the capacity of AT/heparin to interact with thrombin and to inhibit the thrombin-dependent cleavage of fibrinogen (Figure 5d). These results again support the view that long-term interactions with AuECYN do not significantly impair either the global or local active-site structures of thrombin.

CONCLUSIONS

The application of synthetic nanomaterials as exogenous effectors of enzyme structure and catalysis raises interesting new possibilities toward the regulation of biological processes and pathways. Herein, we have prepared two ultrasmall AuNPs displaying net negative charges and virtually identical size distributions but differing in the nature of their surface chemistries: AuMBA, which was passivated with *p*-mercaptobenzoic acid, and AuECYN, which was stabilized with a peptide-based biomimetic coat (Ac-ECYN). We then carried out a systematic investigation of the long-term impact of both AuNPs on the structure and function of human α-thrombin.

We determined that AuMBA and AuECYN had similar apparent binding affinities and kinetics of interactions with thrombin. In addition, we found a remarkable binding selectivity of the AuNPs toward specific protein surface regions. In contrast to AuMBA, which we have previously shown to bind readily to thrombin's exosites 1 and 2, AuECYN interacted primarily with exosite 2. In part, the differential

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001

binding of AuMBA and AuECYN to exosite 1 may be the result of solvation effects.

Nanoparticle complexation at the exosites prompted an allosterically transmitted conformational change in the active site, leading in turn to an allosteric partial inhibition of the enzyme activity. However, we observed that extending the interactions of thrombin with AuMBA over 24 h produced greater and more permanent structural changes to the active-site region when compared to AuECYN. Thrombin was therefore completely and irreversibly inactivated toward the cleavage of synthetic substrates by AuMBA, whereas AuECYN caused only partial inhibition of the activity under otherwise identical conditions; moreover, upon dissociation of the AuECYN−thrombin complex, the original activity was fully restored. It was finally established that long-term interactions with AuECYN did not impair the ability of thrombin to form functional complexes with the physiological substrates fibrinogen and antithrombin.

It is well-known that enzymes can become permanently inactivated by strongly adsorbing onto the surface of large NPs and undergoing denaturation. Here, we have shown that permanent enzyme inactivation can also follow from short-lived interactions with ultrasmall AuMBA under certain conditions, even though the overall structure of the protein might not be markedly affected. This behavior is in agreement with the notion that the catalytic site displays more conformational flexibility than the enzyme as a whole, so that it may be more easily perturbed by external factors.

In summary, the above findings demonstrate that ultrasmall AuNPs may selectively bind to specific protein surface regions through proper design of their surface chemistries. In addition, our findings reveal how the distinct binding modes of different AuNP types may uniquely influence the structure and function of an enzyme's active site (Figure 6). In that regard, AuMBA, which binds to both exosites 1 and 2, can be conceived as a "mild denaturant" of thrombin leading to catalytic dysfunction over time. Conversely, AuECYN, which binds selectively to exosite 2, more closely resembles a proper allosteric effector.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.langmuir.0c01352.

> Quantification of AuNP−thrombin binding affinity by SPR biosensing; dependence of AuNP−thrombin interactions on solution ionic strength; stopped-flow spectroscopy characterization of AuNP−thrombin interactions; CD spectroscopy characterization of AuNP−thrombin interactions; tryptophan fluorescence in samples of thrombin plus AuNPs; FITC fluorescence in samples of FITC−thrombin plus AuNPs; fluorescence quenching of thrombin by AuNPs in the presence of aptamers; inhibition of thrombin activity by AuNPs in the presence of aptamers; inhibition of thrombin activity by AuNPs assessed with a fluorogenic substrate; inhibition of thrombin activity by AuNPs as a function of mixing time; anisotropy measurements of DAPA−thrombin interactions (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Alioscka A. Sousa − *Department of Biochemistry, Federal University of São Paulo, São Paulo, SP 04044-020, Brazil;* orcid.org/0000-0001-7443-5363; Email: alioscka.sousa@unifesp.br

Authors

André L. Lira − *Department of Biochemistry, Federal University of São Paulo, São Paulo, SP 04044-020, Brazil*

Rodrigo S. Ferreira − *Department of Biochemistry, Federal University of São Paulo, São Paulo, SP 04044-020, Brazil*

Maria Luiza V. Oliva − *Department of Biochemistry, Federal University of São Paulo, São Paulo, SP 04044-020, Brazil*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.langmuir.0c01352

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We acknowledge the Brazilian Biosciences National Laboratory (LNBio), CNPEM, Campinas, for their support with the analytical ultracentrifuge, and Dr. Ricardo Torquato for assistance with the SPR measurements. This work was supported by the São Paulo Research Foundation (FAPESP: 2019/04372-6 and 2016/25535-2).

■ REFERENCES

(1) Kopp, M.; Kollenda, S.; Epple, M. Nanoparticle-protein interactions: therapeutic approaches and supramolecular chemistry. *Acc. Chem. Res.* **2017**, *50*, 1383−1390.

(2) Kotov, N. A. Inorganic nanoparticles as protein mimics. *Science* **2010**, *330*, 188−189.

(3) Rodriguez-Quijada, C.; Sánchez-Purrá, M.; de Puig, H.; Hamad-Schifferli, K. Physical properties of biomolecules at the nanomaterial interface. *J. Phys. Chem. B* **2018**, *122*, 2827−2840.

(4) Koide, H.; Yoshimatsu, K.; Hoshino, Y.; Lee, S.-H.; Okajima, A.; Ariizumi, S.; Narita, Y.; Yonamine, Y.; Weisman, A. C.; Nishimura, Y.; Oku, N.; Miura, Y.; Shea, K. J. A polymer nanoparticle with engineered affinity for a vascular endothelial growth factor (VEGF 165). *Nat. Chem.* **2017**, *9*, 715−722.

(5) Chen, M.; Zeng, G.; Xu, P.; Lai, C.; Tang, L. How do enzymes 'meet'nanoparticles and nanomaterials? *Trends Biochem. Sci.* **2017**, *42*, 914−930.

(6) Wu, Z.; Zhang, B.; Yan, B. Regulation of enzyme activity through interactions with nanoparticles. *Int. J. Mol. Sci.* **2009**, *10*, 4198−4209.

(7) Nussinov, R.; Tsai, C.-J. The different ways through which specificity works in orthosteric and allosteric drugs. *Curr. Pharm. Des.* **2012**, *18*, 1311−1316.

(8) Cha, S.-H.; Hong, J.; McGuffie, M.; Yeom, B.; VanEpps, J. S.; Kotov, N. A. Shape-dependent biomimetic inhibition of enzyme by nanoparticles and their antibacterial activity. *ACS Nano* **2015**, *9*, 9097−9105.

(9) Liu, Y.; Winkler, D. A.; Epa, V. C.; Zhang, B.; Yan, B. Probing enzyme-nanoparticle interactions using combinatorial gold nanoparticle libraries. *Nano Res.* **2015**, *8*, 1293−1308.

(10) Breger, J. C.; Ancona, M. G.; Walper, S. A.; Oh, E.; Susumu, K.; Stewart, M. H.; Deschamps, J. R.; Medintz, I. L. Understanding how nanoparticle attachment enhances phosphotriesterase kinetic efficiency. *ACS Nano* **2015**, *9*, 8491−8503.

(11) Deka, J.; Paul, A.; Chattopadhyay, A. Modulating enzymatic activity in the presence of gold nanoparticles. *RSC Adv.* **2012**, *2*, 4736−4745.

(12) Gilles, P.; Wenck, K.; Stratmann, I.; Kirsch, M.; Smolin, D. A.; Schaller, T.; de Groot, H.; Kraft, A.; Schrader, T. High-Affinity

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

Copolymers Inhibit Digestive Enzymes by Surface Recognition. *Biomacromolecules* **2017**, *18*, 1772−1784.
(13) Liu, Y.; Fu, J.; Pan, W.; Xue, Q.; Liu, X.; Zhang, A. Inhibition of thrombin by functionalized C60 nanoparticles revealed via in vitro assays and in silico studies. *J. Environ. Sci.* **2018**, *63*, 285−295.
(14) Sandanaraj, B. S.; Vutukuri, D. R.; Simard, J. M.; Klaikherd, A.; Hong, R.; Rotello, V. M.; Thayumanavan, S. Noncovalent modification of chymotrypsin surface using an amphiphilic polymer scaffold: implications in modulating protein function. *J. Am. Chem. Soc.* **2005**, *127*, 10693−10698.
(15) Stueker, O.; Ortega, V. A.; Goss, G. G.; Stepanova, M. Understanding interactions of functionalized nanoparticles with proteins: a case study on lactate dehydrogenase. *Small* **2014**, *10*, 2006−2021.
(16) You, C.-C.; Agasti, S. S.; De, M.; Knapp, M. J.; Rotello, V. M. Modulation of the catalytic behavior of α-chymotrypsin at monolayer-protected nanoparticle surfaces. *J. Am. Chem. Soc.* **2006**, *128*, 14612−14618.
(17) Lira, A. L.; Ferreira, R. S.; Torquato, R. J.; Oliva, M. L. V.; Schuck, P.; Sousa, A. A. Allosteric inhibition of α-thrombin enzymatic activity with ultrasmall gold nanoparticles. *Nanoscale Adv.* **2019**, *1*, 378−388.
(18) Algar, W. R.; Jeen, T.; Massey, M.; Peveler, W. J.; Asselin, J. Small surface, big effects, and big challenges: Toward understanding enzymatic activity at the inorganic nanoparticle-substrate interface. *Langmuir* **2019**, *35* (22), 7067−7091.
(19) Liu, Y.; Yan, B.; Winkler, D. A.; Fu, J.; Zhang, A. Competitive inhibition mechanism of acetylcholinesterase without catalytic active site interaction: study on functionalized C60 nanoparticles via in vitro and in silico assays. *ACS Appl. Mater. Interfaces* **2017**, *9*, 18626−18638.
(20) Tellechea, E.; Wilson, K. J.; Bravo, E.; Hamad-Schifferli, K. Engineering the interface between glucose oxidase and nanoparticles. *Langmuir* **2012**, *28*, 5190−5200.
(21) Mãnnel, M. J.; Kreuzer, L. P.; Goldhahn, C.; Schubert, J.; Hartl, M. J.; Chanana, M. Catalytically active protein coatings: Toward enzymatic cascade reactions at the intercolloidal level. *ACS Catal.* **2017**, *7*, 1664−1672.
(22) Küchler, A.; Yoshimoto, M.; Luginbühl, S.; Mavelli, F.; Walde, P. Enzymatic reactions in confined environments. *Nat. Nanotechnol.* **2016**, *11*, 409−420.
(23) Chen, M.; Zeng, G.; Xu, P.; Yan, M.; Xiong, W.; Zhou, S. Interaction of carbon nanotubes with microbial enzymes: conformational transitions and potential toxicity. *Environ. Sci.: Nano* **2017**, *4*, 1954−1960.
(24) Wang, Y.; Jonkutė, R.; Lindmark, H.; Keighron, J. D.; Cans, A.-S. Molecular Crowding and a Minimal Footprint at a Gold Nanoparticle Support Stabilize Glucose Oxidase and Boost its Activity. *Langmuir* **2020**, *36*, 37−46.
(25) Nakamoto, M.; Zhao, D.; Benice, O. R.; Lee, S.-H.; Shea, K. J. Abiotic Mimic of Endogenous Tissue Inhibitors of Metalloproteinases: Engineering Synthetic Polymer Nanoparticles for Use as a Broad-Spectrum Metalloproteinase Inhibitor. *J. Am. Chem. Soc.* **2020**, *142*, 2338−2345.
(26) Chen, W.-Q.; Yin, M.-m.; Song, P.-J.; He, X.-H.; Liu, Y.; Jiang, F.-L. Thermodynamics, Kinetics and Mechanisms of Non-competitive Allosteric Inhibition of Chymotrypsin by Dihydrolipoic Acid-Coated Gold Nanoclusters. *Langmuir* **2020**, *36*, 6447−6457.
(27) Vranish, J. N.; Ancona, M. G.; Walper, S. A.; Medintz, I. L. Pursuing the Promise of Enzymatic Enhancement with Nanoparticle Assemblies. *Langmuir* **2018**, *34*, 2901−2925.
(28) Ding, S.; Cargill, A. A.; Medintz, I. L.; Claussen, J. C. Increasing the activity of immobilized enzymes with nanoparticle conjugation. *Curr. Opin. Biotechnol.* **2015**, *34*, 242−250.
(29) Díaz, S. n. A.; Sen, S.; Boeneman Gemmill, K.; Brown III, C. W.; Oh, E.; Susumu, K.; Stewart, M. H.; Breger, J. C.; Lasarte Aragonés, G.; Field, L. D.; et al. Elucidating surface ligand-dependent kinetic enhancement of proteolytic activity at surface-modified quantum dots. *ACS Nano* **2017**, *11*, 5884−5896.
(30) Verespy, S., III; Mehta, A. Y.; Afosah, D.; Al-Horani, R. A.; Desai, U. R. Allosteric partial inhibition of monomeric proteases. Sulfated coumarins induce regulation, not just inhibition, of thrombin. *Sci. Rep.* **2016**, *6*, 24043.
(31) Dokholyan, N. V. Controlling allosteric networks in proteins. *Chem. Rev.* **2016**, *116*, 6463−6487.
(32) Shao, Q.; Hall, C. K. Allosteric effects of gold nanoparticles on human serum albumin. *Nanoscale* **2017**, *9*, 380−390.
(33) Zarschler, K.; Rocks, L.; Licciardello, N.; Boselli, L.; Polo, E.; Garcia, K. P.; De Cola, L.; Stephan, H.; Dawson, K. A. Ultrasmall inorganic nanoparticles: state-of-the-art and perspectives for biomedical applications. *Nanomedicine* **2016**, *12*, 1663−1701.
(34) Lira, A. L.; Ferreira, R. S.; Torquato, R. J.; Zhao, H.; Oliva, M. L. V.; Hassan, S. A.; Schuck, P.; Sousa, A. A. Binding kinetics of ultrasmall gold nanoparticles with proteins. *Nanoscale* **2018**, *10*, 3235−3244.
(35) Ferreira, R. S.; Lira, A. L.; Torquato, R. J.; Schuck, P.; Sousa, A. A. Mechanistic Insights Into Ultrasmall Gold Nanoparticle-Protein Interactions Through Measurement of Binding Kinetics. *J. Phys. Chem. C* **2019**, *123*, 28450−28459.
(36) Boselli, L.; Polo, E.; Castagnola, V.; Dawson, K. A. Regimes of biomolecular ultrasmall nanoparticle interactions. *Angew. Chem., Int. Ed.* **2017**, *56*, 4215−4218.
(37) Huntington, J. A. Thrombin plasticity. *Biochim. Biophys. Acta, Proteins Proteomics* **2012**, *1824*, 246−252.
(38) Malovichko, M. V.; Sabo, T. M.; Maurer, M. C. Ligand binding to anion-binding exosites regulates conformational properties of thrombin. *J. Biol. Chem.* **2013**, *288*, 8667−8678.
(39) Bock, P. E. Thioester peptide chloromethyl ketones: Reagents for active site-selective labeling of serine proteinases with spectroscopic probes. In *Methods in Enzymology*; Elsevier: Amsterdam, Netherlands, 1993; Vol. *222*, pp 478−503.
(40) Sousa, A. A.; Hassan, S. A.; Knittel, L. L.; Balbo, A.; Aronova, M. A.; Brown, P. H.; Schuck, P.; Leapman, R. D. Biointeractions of ultrasmall glutathione-coated gold nanoparticles: effect of small size variations. *Nanoscale* **2016**, *8*, 6577−6588.
(41) Heinecke, C. L.; Ackerson, C. J. Preparation of gold nanocluster bioconjugates for electron microscopy. *Methods Mol. Biol.* **2013**, *950*, 293−311.
(42) Sousa, A. A. A note on the use of steady-state fluorescence quenching to quantify nanoparticle-protein interactions. *J. Fluoresc.* **2015**, *25*, 1567−1575.
(43) Sousa, A. A. Biophysical Characterization of Nanoparticle−Protein Interactions by Fluorescence Quenching Titration: Limitations, Pitfalls, and Application of a Model-Free Approach for Data Analysis. In *Reviews in Fluorescence 2017*; Springer: New York, 2018; pp 53−73.
(44) Rasmussen, T.; Edwards, M. D.; Black, S. S.; Rasmussen, A.; Miller, S.; Booth, I. R. Tryptophan in the Pore of the Mechanosensitive Channel MscS: Assessment of Pore Conformations by Fluorescence Spectroscopy. *J. Biol. Chem.* **2010**, *285*, 5377−5384.
(45) Svitel, J.; Balbo, A.; Mariuzza, R. A.; Gonzales, N. R.; Schuck, P. Combined affinity and rate constant distributions of ligand populations from experimental surface binding kinetics and equilibria. *Biophys. J.* **2003**, *84*, 4062−4077.
(46) Svitel, J.; Boukari, H.; Van Ryk, D.; Willson, R. C.; Schuck, P. Probing the functional heterogeneity of surface binding sites by analysis of experimental binding traces and the effect of mass transport limitation. *Biophys. J.* **2007**, *92*, 1742−1758.
(47) Knittel, L. L.; Schuck, P.; Ackerson, C. J.; Sousa, A. A. Zwitterionic glutathione monoethyl ester as a new capping ligand for ultrasmall gold nanoparticles. *RSC Adv.* **2016**, *6*, 46350−46355.
(48) Knittel, L. L.; Zhao, H.; Nguyen, A.; Miranda, A.; Schuck, P.; Sousa, A. A. Ultrasmall Gold Nanoparticles Coated with Zwitterionic Glutathione Monoethyl Ester: A Model Platform for the Incorporation of Functional Peptides. *J. Phys. Chem. B* **2020**, *124*, 3892−3902.
(49) Pica, A.; Russo Krauss, I.; Parente, V.; Tateishi-Karimata, H.; Nagatoishi, S.; Tsumoto, K.; Sugimoto, N.; Sica, F. Through-bond

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001

effects in the ternary complexes of thrombin sandwiched by two DNA aptamers. *Nucleic Acids Res.* **2017**, *45*, 461−469.
(50) Fiorucci, S.; Zacharias, M. Prediction of protein-protein interaction sites using electrostatic desolvation profiles. *Biophys. J.* **2010**, *98*, 1921−1930.
(51) Ahmad, M.; Gu, W.; Geyer, T.; Helms, V. Adhesive water networks facilitate binding of protein interfaces. *Nat. Commun.* **2011**, *2*, 1−7.
(52) Moyano, D. F.; Ray, M.; Rotello, V. M. Nanoparticle-protein interactions: Water is the key. *MRS Bull.* **2014**, *39*, 1069−1073.
(53) Hassan, S. A. Implicit treatment of solvent dispersion forces in protein simulations. *J. Comput. Chem.* **2014**, *35*, 1621−1629.
(54) Rodier, F.; Bahadur, R. P.; Chakrabarti, P.; Janin, J. Hydration of protein-protein interfaces. *Proteins: Struct., Funct., Genet.* **2005**, *60*, 36−45.
(55) Ulucan, O.; Helms, V. How hydrophilic proteins form nonspecific complexes. *J. Phys. Chem. B* **2015**, *119*, 10524−10530.
(56) Bock, P. E. Active site selective labeling of serine proteases with spectroscopic probes using thioester peptide chloromethyl ketones: demonstration of thrombin labeling using N. alpha.[(acetylthio) acetyl]-D-Phe-Pro-Arg-Ch2Cl. *Biochemistry* **1988**, *27*, 6633−6639.
(57) Lakowicz, J. R. *Principles of Fluorescence Spectroscopy*; Springer: New York, 2006.
(58) Baici, A. Kinetics of enzyme-modifier interactions. In *Selected Topics in the Theory and Diagnosis of Inhibition and Activation Mechanisms*; Springer: New York, 2015.
(59) Zhou, Y.; Qi, X.; Liu, Y.; Zhang, F.; Yan, H. DNA-Nanoscaffold-Assisted Selection of Femtomolar Bivalent Human α-Thrombin Aptamers with Potent Anticoagulant Activity. *ChemBioChem* **2019**, *20*, 2494−2503.
(60) Nesheim, M. E.; Prendergast, F. G.; Mann, K. G. Interactions of a fluorescent active-site-directed inhibitor of thrombin: dansylarginine N-(3-ethyl-1, 5-pentanediyl) amide. *Biochemistry* **1979**, *18*, 996−1003.
(61) Daniel, R. M.; Danson, M. J. Temperature and the catalytic activity of enzymes: A fresh understanding. *FEBS Lett.* **2013**, *587*, 2738−2743.
(62) Tsou, C.-L. Conformational flexibility of enzyme active sites. *Science* **1993**, *262*, 380−382.
(63) Xie, Y.; An, J.; Yang, G.; Wu, G.; Zhang, Y.; Cui, L.; Feng, Y. Enhanced enzyme kinetic stability by increasing rigidity within the active site. *J. Biol. Chem.* **2014**, *289*, 7994−8006.
(64) Zhou, H.; Zhang, X.-H.; Yin, Y.; Tsou, C. Conformational changes at the active site of creatine kinase at low concentrations of guanidinium chloride. *Biochem. J.* **1993**, *291*, 103−107.
(65) Wu, M.; Algar, W. R. Acceleration of proteolytic activity associated with selection of thiol ligand coatings on quantum dots. *ACS Appl. Mater. Interfaces* **2015**, *7*, 2535−2545.
(66) Huntington, J. Molecular recognition mechanisms of thrombin. *J. Thromb. Haemostasis* **2005**, *3*, 1861−1872.
(67) Kurniawan, N. A.; van Kempen, T. H.; Sonneveld, S.; Rosalina, T. T.; Vos, B. E.; Jansen, K. A.; Peters, G. W.; van de Vosse, F. N.; Koenderink, G. H. Buffers strongly modulate fibrin self-assembly into fibrous networks. *Langmuir* **2017**, *33*, 6342−6352.

Downloaded from pubs.acs.org/langd5/article-pdf/36/27/7991/6280804/la0c01352.pdf by HARVARD UNIV user on 09 November 2026

https://dx.doi.org/10.1021/acs.langmuir.0c01352
Langmuir 2020, 36, 7991−8001